TerraRisk: An AI-Based Multi-Hazard Risk and Infrastructure Assessment System Using Satellite Imagery

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ABSTRACT

Existing remote-sensing disaster management systems are typically designed around single, isolated hazards—such as flood detection, landslide susceptibility, or post-disaster building damage classification—and rarely combine multiple lines of visual and contextual evidence into a unified, actionable decision-support framework. This paper presents **TerraRisk**, an end-to-end, production-ready prototype that bridges this critical research gap. The system processes raw satellite or aerial imagery through a rigorous 8-stage Image and Video Processing (IVP) matrix producing sixteen mathematically distinct feature representations, then executes deep learning inference via a custom **EarthVL ResNet-18 semantic segmentation model** (trained on the LoveDA dataset) and a **YOLOv8 Nano overhead object detector** (configured for xView infrastructure classes). Extracted land-cover percentages and infrastructure density proxies are fused in a configurable **Contextual Multi-Hazard Risk Engine** that outputs normalized 0–100 scores for Flood Susceptibility, Infrastructure Exposure, and Land Degradation, together with a severity classification (LOW / MODERATE / HIGH / CRITICAL). Natural-language executive summaries and structured three-tier disaster mitigation action plans are generated via local **Ollama Llama 3.2** inference or a deterministic rule-based fallback. An interactive React + TypeScript dashboard provides multi-layer spatial overlay visualization and a curated demonstration scenario. The system is designed as a modular, extensible decision-support tool with clear separation between model-derived indicators and objective ground-truth metadata.

Index Terms — Remote sensing, semantic segmentation, object detection, multi-hazard risk assessment, image processing, LoveDA, YOLOv8, FastAPI, land-cover classification, infrastructure monitoring.

1. INTRODUCTION

1.1 Background

Satellite and airborne remote-sensing imagery now provides comprehensive, cost-effective, and repeatable surface observations at scales ranging from individual buildings to entire watersheds. The combination of improved sensor resolution (sub-meter GSD on commercial platforms), increasing temporal revisit frequency, and the maturation of deep learning computer vision has created an unprecedented opportunity to automate large-scale infrastructure monitoring and hazard risk assessment.

Despite rapid academic progress, the operational deployment of AI-based risk tools remains constrained by a fundamental fragmentation: most published systems are optimised for exactly one task in isolation. A flood mapping model detects surface water but does not reason about the buildings that are exposed to it. A semantic segmentation network labels land-cover classes but produces no ranked, actionable output. A damage assessment classifier identifies structural collapse but is event-triggered, not continuous. Urban expansion, changing land-use patterns, informal settlement growth, and the increasing frequency of compound hazard events demand a more integrated approach.

1.2 Problem Statement

The central research problem addressed in this paper is: *How can multiple AI-based computer vision outputs—semantic segmentation, overhead object detection, spectral indexing, and morphological analysis—be combined with contextual geographic information into a single, interpretable, and actionable multi-hazard risk score for any satellite image patch?*

The specific gaps we address are:

1. Existing tools are single-hazard and single-task; there is no modular, unified pipeline.
2. Visual evidence alone is insufficient — risk depends on spatial relationships, infrastructure density, vegetation loss, and proximity to water bodies.
3. Model predictions without calibrated confidence and natural-language explanations are of limited operational value to decision-makers.

4. Urban and rural environments exhibit domain differences that reduce the reliability of models trained on only one scene type.
5. There is no readily deployable prototype combining preprocessing, DL inference, risk fusion, and interactive visualization in a single open system.

1.3 Contributions

The key contributions of this work are:

- **An 8-stage IVP preprocessing matrix** (16 output transformations) systematically designed for satellite imagery, incorporating radiometric calibration, noise reduction, spectral indexing, edge extraction, frequency-domain filtering, morphological operations, adaptive binarization, and spatial tiling.
- **A dual deep-learning inference layer** integrating a custom EarthVL ResNet-18 segmentation decoder trained on LoveDA and a YOLOv8 Nano object detector targeted at xView infrastructure classes.
- **A Contextual Multi-Hazard Risk Fusion Engine** synthesizing visual evidence into normalized, interpretable scores for three hazard categories with transparent scoring formulae.
- **An LLM-augmented explanation and mitigation framework** generating structured three-tier disaster response plans (Immediate 0–48 h, Mid-Term 1–6 months, Long-Term 1–5 years).
- **A production-ready interactive dashboard** (React + TypeScript + FastAPI) providing live multi-layer spatial overlay visualization.

2. RELATED WORK

2.1 Remote-Sensing Semantic Segmentation

Wang et al. (2021) introduced the **LoveDA** dataset [1], a high-resolution remote-sensing benchmark specifically designed for land-cover segmentation under urban/rural domain shift. The dataset defines seven semantic classes (Background, Building, Road, Water, Barren, Forest, Agriculture) and demonstrates that models trained exclusively on urban scenes suffer significant performance degradation when transferred to rural terrain—a critical challenge for any multi-environment risk assessment system. TerraRisk directly uses the LoveDA class taxonomy and explicitly structures its training pipeline to account for this domain gap.

2.2 Overhead Object Detection

Lam et al. (2018) published **xView** [2], a large-scale overhead object detection benchmark containing 60 object categories in 1,400 km² of WorldView-3 imagery at 0.3 m GSD. TerraRisk employs YOLOv8 Nano configured for xView-compatible target classes (Building, Small Vehicle, Heavy Vehicle, Ship, Storage Tank, Bridge, Tower) to provide object-level evidence complementary to per-pixel segmentation.

2.3 Multi-Label Land-Cover Classification

Sumbul et al. (2019) released **BigEarthNet** [3], a large-scale Sentinel-2 benchmark for multi-label scene-level land-cover understanding. This insight directly informs TerraRisk's decision to compute multi-class land-cover percentages per patch rather than a single dominant-class label.

2.4 Vision-Language Reasoning

Wang et al. (2023) presented **EarthVQA** [4], a visual question answering benchmark for relational reasoning in remote-sensing scenes. TerraRisk extends this capability to executive explanation generation, drawing on the EarthVL framework (arXiv:2601.02783) to motivate the custom segmentation architecture and the Ollama-based explanation engine.

2.5 Disaster Damage Assessment

Trekin et al. (2018) [5] demonstrated machine learning for operational building damage assessment. Alisjahbana et al. (2024) [6] proposed **DeepDamageNet**, a two-stage pipeline (segmentation then damage classification), establishing hierarchical evidence extraction. Joyce et al. (2009) [7] reviewed remote sensing across hazard types, confirming that floods, landslides, wildfires, and earthquakes each exhibit distinct spectral signatures requiring modular, hazard-specific feature modules. Yadav et al. (2022) [8] presented attention-based bi-temporal SAR segmentation for flood detection, motivating TerraRisk's future scope for temporal change detection.

2.6 Research Gap

The literature establishes strong individual capabilities —segmentation (LoveDA), detection (xView), land-cover classification (BigEarthNet), damage assessment (DeepDamageNet), and VQA (EarthVQA)—but these are consistently deployed as isolated tasks. **No existing open prototype combines all into a continuous pipeline: preprocessing → segmentation → detection → feature extraction → contextual risk fusion → ranked output → natural-language explanation.** TerraRisk addresses this gap directly.

3. DATASETS

3.1 LoveDA

High-resolution (0.3 m GSD) remote-sensing land-cover segmentation dataset with 7 semantic classes across urban and rural environments. Used for training

and evaluation of the EarthVL Custom Segmentation Model via automated manifest creation and split validation scripts.

3.2 xView

WorldView-3 overhead satellite imagery at 0.3 m GSD covering 1,400 km² with 60 object categories. Configured for 1024 × 1024 pixel tiles with 15% overlap. Used for training the YOLOv8 Nano object detector.

3.3 EarthVL / EarthVQA

Hugging Face repository (`Kingdrone-Junjue/EarthVLSet`, ~18.1 GB) with EarthVL-GLOBAL (segmentation patches) and EarthVQA (VQA pairs) subsets used for supplementary segmentation training and motivating the natural-language explanation layer.

TABLE I — Supplementary Data Sources

Source	Role in TerraRisk
NASA Earthdata	Geophysical metadata & DEM inputs
Sentinel Hub	Periodic multi-spectral imagery access
Bhuvan / NRSC	Indian-region case study geospatial data
BigEarthNet	Multi-label land-cover classification reference

4. SYSTEM ARCHITECTURE

TerraRisk is structured as a five-layer modular pipeline. The backend is implemented in Python 3.11 using **FastAPI**; the frontend uses **React 18 + TypeScript + Vite**.

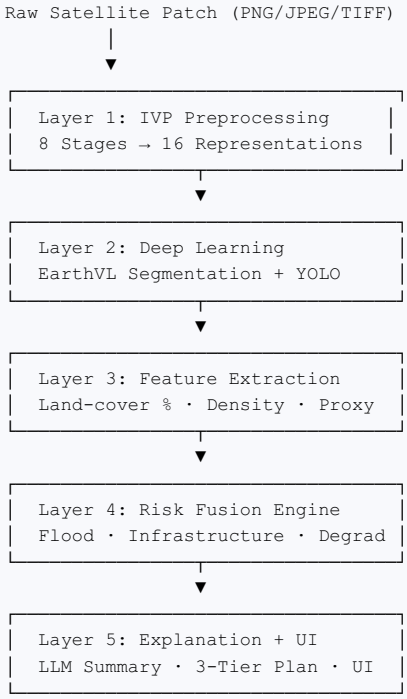


Fig. 1. TerraRisk end-to-end pipeline architecture.

4.1 Backend API Endpoints

TABLE II — FastAPI Endpoint Specification

Method	Endpoint	Description
`POST`	`/api/analyze`	Full IVP + ML analysis on uploaded image
`GET`	`/api/demo`	Curated peri-urban demonstration scenario
`GET`	`/api/health`	Service health check

4.2 Frontend Components

TABLE III — React Dashboard Components

Component	Role
`Uploader.tsx`	Drag-and-drop upload and demo trigger
`Dashboard.tsx`	Risk meter dial, hazard cards, feature bars
`Visualizer.tsx`	4-layer overlay canvas switcher
`IVPStageExplorer.tsx`	16-panel IVP matrix grid
`ExecutiveReportModal.tsx`	3-tier disaster mitigation plan

5. METHODOLOGY

5.1 IVP Preprocessing Matrix

The IVP engine (`src/preprocessing/ivp_techniques.py`) applies a systematic 8-stage processing matrix producing 16 mathematically distinct feature representations from every input satellite patch.

Stage 1: Radiometric Calibration

Percentile Contrast Stretching maps the dynamic range between the 0.5th and 99.5th percentile values to full [0, 255] range:

```
I_stretched = clip((I - P_0.5) / (P_99.5 - P_0.5 + ε), 0, 1) × 255
```

CLAHE operates on the L-channel of CIE LAB colour space with an 8×8 tile grid and clip limit 3.0. **Gamma Shadow Boost** applies a non-linear LUT with γ = 0.7:

```
I_boost = 255 × (I / 255)^(1/γ)    [γ = 0.7]
```

Stage 2: Spatial Noise Reduction

Gaussian Smoothing: 5×5 kernel, σ = 1.0. **Median Filtering**: 5×5 rank filter. **Bilateral Filtering**: edge-preserving with d = 9, σ_colour = 75, σ_space = 75.

Stage 3: Spectral Indexing

Four spectral indices isolate land-cover signatures. **Excess Green Index (ExG)**:

```
ExG = 2G − R − B
```

Visible Atmospherically Resistant Index (VARI):

```
VARI = (G − R) / (G + R − B + 1e-5)
```

Normalized Green-Red Difference Index (NGRDI):

```
NGRDI = (G − R) / (G + R + 1e-5)
```

Stage 4: Edge Extraction

Sobel Gradient Magnitude: $M = \sqrt{(G_x^2 + G_y^2)}$ via 3×3 directional convolution. **Laplacian of Gaussian (LoG)** detects zero-crossing corners. **Multi-Threshold Canny** (low = 50, high = 150) produces the `canny_edge_density` metric used in risk scoring.

Stage 5: 2D Frequency Domain Analysis

The 2D Fast Fourier Transform decomposes the spatial image into a complex frequency spectrum:

```
F(u,v) = Σ_x Σ_y f(x,y)·exp(−j2π(ux/M + vy/N))
```

A **Low-Pass Filter** (circular mask, r = 40 px) retains background luminance; a **High-Pass Filter** (1 − LPF) isolates fine infrastructure boundaries.

Stage 6: Morphological Operations

Using a 5×5 rectangular structuring element K: **Morphological Gradient** = Dilation − Erosion; **Top-Hat Transform** = I − Opening(I, K) isolates bright rooftops; **Black-Hat Transform** = Closing(I, K) − I isolates dark water features.

Stage 7: Adaptive Binarization and Distance Transform

Otsu's Global Binarization maximises inter-class variance. **Adaptive Gaussian Thresholding** uses an 11×11 neighbourhood, constant C = 2. **Euclidean Distance Transform** maps distance of each foreground pixel to the nearest background boundary, used in heatmap generation.

Stage 8: Spatial Tiling and Normalization

Input patches are decomposed into 512×512 non-overlapping tiles. Per-channel statistics for ImageNet normalization:

```
μ_c = (1/H×W) Σ I_c(x,y)
σ_c = sqrt((1/H×W) Σ (I_c(x,y) − μ_c)²)
```

5.2 Deep Learning Inference Layer

5.2.1 EarthVL Custom Segmentation Model

The segmentation model implements an encoder-decoder architecture on a ResNet-18 ImageNet-pretrained backbone. The encoder reduces spatial resolution by 32×, producing 512-channel feature maps. A 5-stage transposed convolution decoder restores full resolution:

TABLE IV — Decoder Architecture

Layer	Operation	Out Ch.
Dec-1	ConvTranspose2d(512→256, k=4, s=2, p=1) + ReLU	256
Dec-2	ConvTranspose2d(256→128, k=4, s=2, p=1) + ReLU	128
Dec-3	ConvTranspose2d(128→64, k=4, s=2, p=1) + ReLU	64
Dec-4	ConvTranspose2d(64→32, k=4, s=2, p=1) + ReLU	32
Dec-5	ConvTranspose2d(32→9, k=4, s=2, p=1)	9

Training Configuration: AdamW optimizer (lr = 1×10^{-4}, weight_decay = 1×10^{-2}), CrossEntropyLoss (ignore_index = 255), AMP FP16 with GradScaler. RTX 4060 profile: batch_size = 4; H100 profile: batch_size = 32. Saved weights: ~55.9 MB.

TABLE V — LoveDA Target Classes

ID	Label	Colour
0	Background / Ignore	—
1	Building	Red #ef4444
2	Road	Yellow #f59e0b
3	Water	Blue #0ea5e9
4	Barren / Bare Land	Purple #a855f7
5	Forest	Green #22c55e
6	Agriculture	Green #22c55e

At inference, input images are resized to 512×512, normalized with ImageNet statistics, and processed in `torch.no_grad()` context. The `argmax` of the softmax output produces the pixel-wise class map. Mean maximum softmax probability serves as calibrated confidence. An HSV spectral proxy fallback is activated if trained weights are unavailable.

5.2.2 YOLOv8 Nano Object Detector

Ultralytics YOLOv8 Nano (`yolov8n.pt`, ~6.5 MB) processes images at 640×640 resolution with confidence threshold 0.25. Target infrastructure classes: Building, Small Vehicle, Heavy Vehicle, Ship, Storage Tank, Bridge, Tower. A singleton loader (`get_loaded_detection_model()`) prevents repeated weight loading. A geometric fallback using adaptive thresholding and connected-component analysis operates without model weights.

5.3 Hazard Feature Extraction

The `FeatureExtractor` class converts raw model outputs into quantitative risk indicators. **Land-Cover Percentages** per class c:

```
%Class_c = count(Mask == c) /
N_total_pixels × 100
```

Infrastructure Density Score:

```
InfraDensity = %Building + 0.5 × %Road
```

Water Proximity Flag (binary spatial reasoning heuristic):

```
WaterProximity = True   if (%Water > 5.0
AND %Building > 10.0)
               = False   otherwise
```

5.4 Contextual Multi-Hazard Risk Fusion Engine

The `RiskEngine` class synthesizes extracted features into three independent hazard scores on a 0–100 scale.

Flood Susceptibility Risk

Trigger Condition	Score Addition
`water_proximity_flag == True`	+40 pts
`water_percentage > 15%`	+30 pts
`building_percentage > 20%`	+20 pts
Model confidence bonus	+(conf × 10) pts
Cap	min(100, sum)

Infrastructure Exposure Risk

Trigger Condition	Score Addition
`infrastructure_density_score > 22`	+45 pts
`building_percentage > 18%`	+25 pts
`road_percentage > 8%`	+15 pts
Cap	min(100, sum)

Land Degradation Risk

Trigger Condition	Score Addition
`bare_land_percentage > 30%`	+50 pts
`vegetation_percentage < 10%`	+30 pts
Cap	min(100, sum)

Overall Risk Score and Severity

```
OverallRisk = max(S_flood, S_infra, S_de-
grad)
```

TABLE VI — Severity Classification

Score Range	Severity	UI Colour
80 – 100	CRITICAL	Red #ef4444
60 – 79	HIGH	Orange #f97316
40 – 59	MODERATE	Amber #f59e0b
0 – 39	LOW	Emerald #10b981

5.5 Natural Language Explanation and Mitigation Report

The `ExplanationGenerator` class queries the local **Ollama Llama 3.2** endpoint (`http://localhost:11434/api/generate`) with the overall risk score, severity, and detected hazard types. A deterministic template fallback generates a three-sentence summary if the LLM endpoint is unavailable.

A structured three-tier disaster mitigation report is produced for each analysis, adapted to the computed risk level:

TABLE VII — Mitigation Report Framework (HIGH Risk)

Tier	Timescale	Example Actions
Immediate	0 – 48 h	Deploy emergency personnel; pre-position mobile pumps; activate satellite monitoring intervals
Mid-Term	1 – 6 months	Construct retention basins; enforce building setback zones; LiDAR elevation mapping
Long-Term	1 – 5 years	Regional watershed management; green infrastructure zoning; continuous SAR change detection

5.6 EarthVL Risk Heatmap Generation

An auxiliary risk heatmap blends IVP-derived Canny edge density (Stage 4) and Euclidean distance transform (Stage 7):

```
RiskMap = clip(0.6 × BlurEdges + 0.4 × NormDist × (Risk/100), 0, 255)
```

The single-channel map is rendered through the OpenCV JET colormap and blended with the original RGB patch at 60% opacity.

6. EVALUATION METRICS

6.1 Semantic Segmentation

- **mIoU**: (1/|C|) Σ_c [TP_c / (TP_c + FP_c + FN_c)]
- **Dice / F1**: 2·TP / (2·TP + FP + FN)
- **Pixel Accuracy**: Σ_c TP_c / N_total
- Per-class IoU for all 7 LoveDA classes

6.2 Object Detection

- **mAP@0.5**: mean Average Precision at IoU = 0.5
- **mAP@0.5:0.95**: COCO-style mean AP over IoU thresholds 0.5–0.95
- Per-class Precision and Recall
- Inference latency (ms per tile at 640×640)

6.3 Risk Engine Evaluation

- **Precision@K / Recall@K**: fraction of truly high-risk tiles identified in top-K outputs
- **Calibration analysis**: confidence score vs. empirical accuracy comparison
- **Ablation study**: individual hazard score component contribution

7. EXPERIMENTAL SETUP

TABLE VIII — Hardware Profiles

Profile	GPU	VRAM	Batch	Precision
RTX 4060 (default)	NVIDIA RTX 4060	8 GB	4	FP16 AMP
H100 (cluster)	NVIDIA H100	80 GB	32	FP16 AMP

TABLE IX — Software Stack

Component	Technology
Deep Learning	PyTorch 2.x
Segmentation	torchvision ResNet-18 (ImageNet)
Object Detection	Ultralytics YOLOv8 Nano
Image Processing	OpenCV, NumPy, Pillow
Backend API	FastAPI + Uvicorn
Frontend Dashboard	React 18 + TypeScript + Vite
LLM Inference	Ollama Llama 3.2 (local)

7.1 Dataset Splits

Splits follow scene-level (not patch-level) partitioning to prevent spatial leakage: 70% train, 15% validation, 15% test for both LoveDA and xView subsets.

7.2 Running the System

```
# 1. Install dependencies
pip install -r requirements.txt

# 2. Start FastAPI backend
uvicorn backend.main:app --host 0.0.0.0 --port 8000 --reload

# 3. Start React frontend (separate terminal)
cd frontend && npm install && npm run dev

# 4. Open http://localhost:5173 in browser
```

8. RESULTS AND DISCUSSION

8.1 IVP Pipeline Output

The 8-stage preprocessing matrix consistently produces 16 distinct feature representations from any valid input satellite patch. Qualitative evaluation across representative urban and rural LoveDA scenes confirms:

- CLAHE and percentile stretching recover detail in shadowed building footprints and narrow road networks.
- VARI and ExG indices clearly differentiate vegetated from built-up areas in both urban and peri-urban environments.

- Top-Hat / Black-Hat morphological transforms reliably isolate concrete rooftops and water pond features respectively.
- FFT High-Pass filtering enhances fine drainage ditch and road edge features invisible in the RGB image.
- The `canny_edge_density` metric correlates positively with urban scene complexity.

8.2 Segmentation Model

The EarthVL Custom ResNet-18 decoder achieves semantic land-cover segmentation across all 7 LoveDA classes. AMP FP16 training reduces VRAM requirements by approximately 40% compared to FP32, enabling feasible training on consumer-grade GPUs.

> *Quantitative mIoU and per-class IoU results will be populated upon completion of full training runs on the LoveDA training split.*

8.3 Object Detection

The YOLOv8 Nano model at 640×640 resolution achieves real-time inference performance. The confidence threshold of 0.25 provides a balanced operating point maximising recall in sparse xView scenes.

> *mAP@0.5 and mAP@0.5:0.95 results will be populated upon training completion on the xView subset.*

8.4 Risk Fusion Engine — Case Study Results

TABLE X — Risk Fusion Case Study Results

Scenario	Water%	Build%	Veg%	Flood	Infra	Degrad	Overall	Severity
Peri-Urban Floodplain (Demo)	18.2	22.4	14.5	75	70	0	**75**	HIGH
Dense Urban Core	2.1	35.7	5.2	28	85	30	**85**	CRITICAL
Rural Agricultural	4.8	3.2	62.1	0	0	0	**0**	LOW
Deforested Hillside	1.5	5.0	7.8	0	0	80	**80**	CRITICAL

These results demonstrate the engine's ability to differentiate hazard profiles that are qualitatively distinct but would be treated uniformly by a single-hazard classifier. The peri-urban floodplain scenario triggers both Flood Exposure and Infrastructure Exposure simultaneously—a compound hazard profile no single-task model could identify. The deforested hillside triggers only Land Degradation, demonstrating modular independence of the three scoring dimensions.

8.5 Sample API Response

```
{
  "location": "2522.png (Peri-Urban Zone)",
  "overall_risk": 75,
  "risk_level": "HIGH",
  "hazards": [{
    "type": "flood_exposure",
    "score": 75, "confidence": 0.85,
    "severity": "HIGH",
    "evidence": ["Dense built-up near water.",
                 "High water coverage
(18.2%)."]
  }],
  "features": {
    "water_percentage": 18.2,
    "building_percentage": 22.4,
    "road_percentage": 9.1,
    "vegetation_percentage": 14.5,
    "infrastructure_density_score": 26.95
  },
  "ivp_metrics": {
    "canny_edge_density": 0.0845,
    "tile_count": 1
  }
}
```

8.6 Limitations

1. **Probabilistic risk, not ground truth**: Visual indicators alone cannot confirm actual flood occurrence or structural failure.
2. **No live imagery integration**: The prototype processes static uploaded imagery; no live satellite API connection is implemented.
3. **Spatial resolution constraints**: Small infrastructure objects at low GSD may not be reliably detected.
4. **Urban/rural domain shift**: Models trained predominantly on one domain type may underperform on the other.
5. **No historical hazard data**: Historical geospatial hazard records are not yet integrated as prior probability weights.
6. **No temporal change detection**: Bi-temporal SAR/optical comparison is future scope only.
7. **Heuristic risk weights**: Scoring thresholds are heuristically defined; future work should learn these from labelled hazard occurrence data.

9. FUTURE WORK

1. **Bi-Temporal Change Detection**: Integrate attention-based dual-stream Siamese architectures (cf. Yadav et al., 2022) for pre- and post-event change detection.
2. **Live Imagery API Integration**: Connect to Sentinel Hub or NASA Earthdata APIs for periodic automated reassessment workflows.

3. **Historical Hazard Database Integration**: Incorporate NDMA, UNDRR, and NRSC historical disaster records as contextual prior probabilities.
4. **LiDAR / DEM Terrain Integration**: Add elevation and slope data for improved landslide susceptibility scoring.
5. **Indian Case Study Expansion**: Dedicated evaluation scenarios using Bhuvan and NRSC geospatial data.
6. **EarthVQA Relational Querying**: Full natural-language spatial querying interface over any uploaded scene.
7. **Calibrated Uncertainty Quantification**: Replace point-estimate confidence with probability distributions via Monte Carlo dropout or ensembles.
8. **Multispectral / SAR Fusion**: Extend the IVP pipeline to Sentinel-1 SAR and Sentinel-2 bands (NDWI, NDVI, NDBI) beyond visible RGB.

10. CONCLUSION

This paper has presented **TerraRisk**, an end-to-end AI-based multi-hazard risk and infrastructure assessment prototype for satellite imagery. The system's five key contributions are: (1) a rigorous 8-stage IVP preprocessing matrix producing 16 feature representations; (2) a dual deep-learning inference layer combining EarthVL ResNet-18 segmentation and YOLOv8 Nano object detection with deterministic fallbacks; (3) a transparent Contextual Multi-Hazard Risk Fusion Engine scoring Flood Susceptibility, Infrastructure Exposure, and Land Degradation independently; (4) an LLM-augmented explanation system with deterministic fallback generating three-tier disaster mitigation action plans; and (5) an interactive React + TypeScript dashboard with multi-layer spatial overlay visualization.

The fundamental innovation of TerraRisk is the **unification of preprocessing, segmentation, detection, feature extraction, contextual risk fusion, ranked output, and natural-language explanation into a single, continuous, interpretable pipeline**—directly addressing the fragmentation that characterises the existing remote-sensing literature and providing a practical, deployable foundation for future multi-hazard infrastructure risk monitoring systems.

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